

Program: Diploma in Food Processing Technology	
Course Code: 4424	Course Title: Food Packaging Technology
Semester: 4	Credits: 2
Course Category: Program Core	
Periods per week: 3 (L: 2T:1 P:0)	Periods per semester: 45

Course Objectives:

1. To study the different materials used for packing and related food products
2. To understand the principles of different packaging systems
3. To impart knowledge on properties of various packaging materials

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of food packaging material	Food packaging technology

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the different uses of packaging materials its manufacturing process and properties	10	Understanding
CO2	Describe the principle and working of different packaging systems	13	Understanding
CO3	Evaluate the physical and barrier properties of different packaging materials	12	Understanding
CO4	Identify the criteria to design packaging material based on type of foods	10	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Stronglymapped,2-Moderatelymapped,1-Weaklymapped

CourseOutline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the different uses of packaging materials its manufacturing process and properties		
M1.01	What are packaging materials, functions of packaging	4	Understanding
M1.02	Review Cushioning materials, types and its characteristics	3	Understanding
M1.03	Illustrate the Properties of glass and its uses, advantages and disadvantages	3	Understanding
Contents: Definitions, objectives, and types of packaging materials; functions of package. Cushioning materials; Paper as food packaging material, Glass: properties, uses advantages & disadvantages			
CO2	Describe the principle and working of different packaging systems		
M2.01	Explain Properties of plastic materials -LDPE, PP, BOPP	4	Understanding
M2.02	Explain Properties of plastic materials -PET, Polyester	4	Understanding
M2.03	Discuss about Co extrusion and lamination techniques	4	Understanding
	Series Test –I	1	
Contents: properties of different types of plastic materials : LDPE, PP, BOPP, PET, Polyester materials, Co extrusion and lamination techniques			

CO3	Evaluate the physical and barrier properties of different packaging materials		
M3.01	Introduce CAP & MAP, vacuum packaging	4	Understanding
M3.02	Explain the Principles, working set up of aseptic packaging	4	Understanding
M3.03	Describe the Principles, working set up of retort packaging	4	Understanding
Contents: Concept, principle and process of MAP and CAP, Vacuum packaging, aseptic packaging and retort packaging,			
CO4	Identify the criteria to design packaging material based on type of foods		
M4.01	Identify the Packaging materials used for and criteria for selection of packaging materials for dairy products	3	Understanding
M4.02	Outline Packaging materials used for and criteria for selection of packaging materials for fruits and vegetable products	3	Understanding
M4.03	Identify Packaging materials used for and criteria for selection of packaging materials for meat and meat products	3	Understanding
	Series Test–II	1	
contents: Packaging materials used for and criteria for selection of packaging materials for the following foods; fruits and vegetable products; meat and poultry products, dairy products.			

Text/Reference

T/R	Book Title/Author
R1	Robertson, G.L. Food Packaging: Principles and Practice, 2nd Edition. Taylor & Francis, 2006.
R2	S. Sacharow and R.C. Griffin, Principles of Food Packaging, AVI Pub. Co., 1988
R3	Han, Jung H. "Innovations in Food Packaging". Elsevier, 2005.
R4	Ahvenainen, Raija. "Novel Food Packaging Techniques". Wood Head Publishing, 2003
R5	M. Mahadeviah and R.V. Gowramma Food Packaging Materials
R6	Paine E.A, "Fundamentals of packaging

Online Resources

Sl. No	Website Link
1	https://www.industrialpackaging.com/blog/what-is-modified-atmosphere-packaging
2	https://foodslord.com/using-map-and-cap-packaging-methods-in-food-industries/
3	https://www.industrialpackaging.com/blog/what-are-packaging-materials